

A blue trapezoidal graphic element on the left side of the title.

# FA report of Ghost Touch (NDF)

JUNGLIM G156HAB03.2

# Problem Statement



- General Information

|               |                    |                      |                  |
|---------------|--------------------|----------------------|------------------|
| Brand/Project | JUNGLIM            | Defect Phenomenon    | Ghost Touch(NDF) |
| ES            |                    | Defect Rate or Q' ty | 1 pc             |
| AUO Model     | G156HAB03.2        | Defect Station       | FIELD REJECT     |
| AUO S/N       | 5Y5B4VJ9KTZZZS0100 | RMA No               |                  |
| Beol/Module   |                    | Project Stage        | MP               |
| Week Code     | 25/27              | Issue Date           | 2026.04.30       |

- Defect Panel S/N Picture



# Problem Statement

- Describe the phenomenon at customer side

1. The customer reported that a Ghost Touch phenomenon occurred during field use.
2. Intermittent touch input is detected in the area marked in red even when no physical touch is applied.
3. JUNGLIM also conducted reproduction verification; however, the reported symptom could not be reproduced.



Customer Reported Symptom

**NUO**

No abnormality was found on the panel appearance. No cosmetic defect, damage, or contamination was observed on either front or rear side.



**NUO**

AUO verified the TP condition using the TP inspection program for the TP NG issue reported by the customer, and the unit was confirmed to be normal with no abnormality detected.

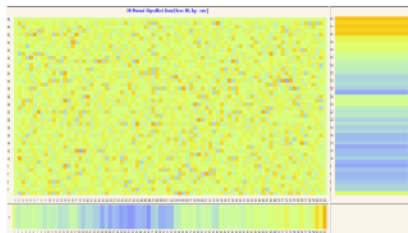
|             |       |      |
|-------------|-------|------|
| <b>PASS</b> | PASS  | 1    |
|             | FAIL  | 0    |
|             | TOTAL | 1    |
|             | YIELD | 100% |

# Root Cause Analysis

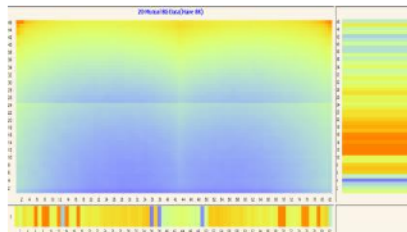


## Step 3: Log Verification:

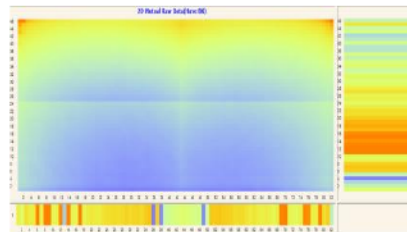
For detailed TP data analysis, CDC logs were collected and analyzed. The analysis results confirmed that all data were within the normal range. The collected log data are provided below for reference.



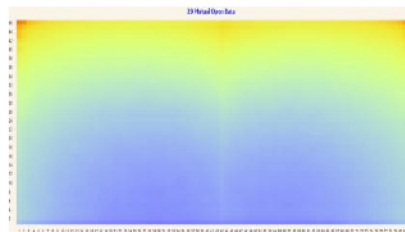
CDC Mutual SignalEnd Data



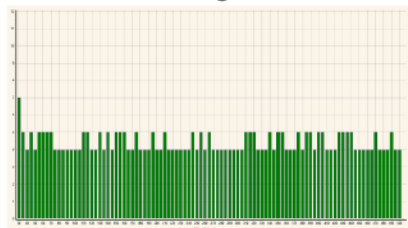
2D Mutual BG Data(Have BK)



2D Mutual Raw Data(Have BK)



2D Mutual Open Data

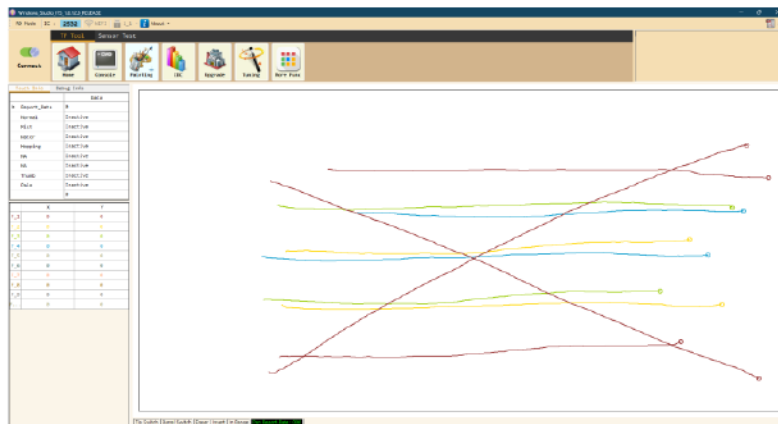


Frequency Noise

# Root Cause Analysis

## Step 4: Drawing Touch Verification:

During drawing and multi-touch operation, normal touch functionality was confirmed with no interruption or malfunction observed.



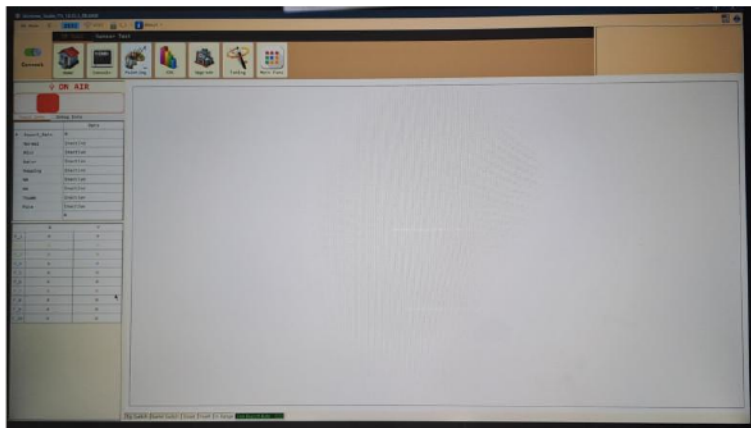
Drawing and Multi-Touch Test

# Root Cause Analysis

## Step 5: Ghost Touch Verification:

A painting program was used to verify whether any Ghost Touch phenomenon occurred, and no abnormality was observed.

A 96-hour recording test was conducted under both room temperature and 40°C conditions for 48 hours each.



|    | A                                             | B           | C    | D    | E    | F    | G    | H    | I | J | K |  |  |  |  |  |
|----|-----------------------------------------------|-------------|------|------|------|------|------|------|---|---|---|--|--|--|--|--|
| 1  | ILITEK Test Studio Utility : Paint V.1.0.12.3 |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 2  | =====                                         |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 3  | IC Type                                       | =ILI2532    |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 4  | Protocol Type                                 | =V6_HID     |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 5  | Protocol Version                              | =6.0.8      |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 6  | Firmware                                      | 0x00        | 0x01 | 0x05 | 0x00 | 0x00 | 0x00 | 0x04 |   |   |   |  |  |  |  |  |
| 7  | X resolution                                  | =16384      |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 8  | Y resolution                                  | =16384      |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 9  | X Channel                                     | =82         |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 10 | Y Channel                                     | =48         |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 11 | Support Fingers                               | =10         |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 12 | Support Touch Keys                            | =0          |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 13 | Key Mode                                      | =NO_Key     |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 14 | Pen Type                                      | =Disable    |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 15 | CommonFlow Ver.                               | =0x00000205 |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 16 | Report Ver.                                   | =0.0.3      |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 17 | =====                                         |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 18 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 19 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 20 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 21 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 22 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 23 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 24 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 25 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 26 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 27 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |
| 28 |                                               |             |      |      |      |      |      |      |   |   |   |  |  |  |  |  |

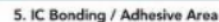
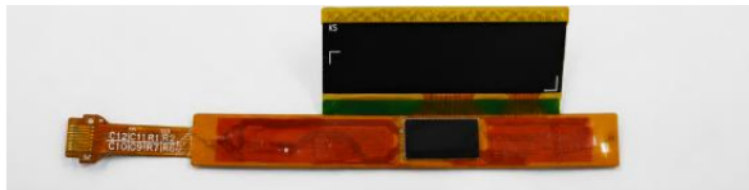
If any touch event had occurred  
data was expected to be recorded

If any touch event had occurred, the corresponding data was expected to be recorded; however, no related data was found in the recorded log.



**NUO**

Surface condition and solder points were inspected using CCD equipment, and no abnormality was observed.



# Root Cause Analysis



## Step 7: TP IC Electrical Status Check

The PCB of the returned sample was maintained under room-temperature conditions, and the electrical status of the TP IC was checked by voltage drop measurement.

The returned sample showed a voltage drop of approximately **0.587 V**, which was consistent with the normal reference condition.

Additionally, a 48-hour aging test at 80°C was conducted on the TP and Touch Controller Board assembly. No Ghost Touch phenomenon was observed during the test, and the reported symptom could not be reproduced.



Good IC voltage drop measurement = 0.587DCV

# Conclusion



## 1. Visual / Hardware Inspection

No abnormality was observed on the panel appearance, PCB surface, solder points, or TP controller board.

## 2. TP Functional / Signal Verification

TP function, drawing operation, multi-touch operation, CDC data, and noise data were confirmed to be within the normal range.

## 3. Ghost Touch Verification

The reported Ghost Touch symptom was not reproduced during Paint Program verification or the 96-hour recording test.

## 4. TP IC Electrical Status Check

The TP IC voltage drop was consistent with the normal reference condition, and no Ghost Touch phenomenon was reproduced during the 48-hour aging test at 80°C.

## 5. Final Judgment

The returned sample is judged as NDF. Continuous monitoring will be maintained for similar future cases.

亮麗創新  曼妙生活

Bright Innovation · Amazing Life



**NUO**

*Innovating Life*